Efficient QLoRA Fine-Tuning of Qwen2.5-VL for Multi-Class Tomato Disease Classification and Cross-Domain Evaluation

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Abstract. QLoRA fine-tuning on Qwen2.5-VL-7B-Instruct for tomato disease and pest classification into 11 classes is studied in this work. Checks for integrity, SHA-256 exact deduplication, and eliminating conflicting cross-class duplicates reduced 7,095 images down to 6,545, of which 5,247 were used for training, 649 for validation, and 649 for testing. The model was equipped with 4-bit NF4 double quantization, BF16 computations, LoRA adapters with rank-64 on language-model attention projections, a frozen vision tower, and answer-only supervision. Training time on one NVIDIA RTX 4500 Ada 24-GB GPU was 192.7 minutes with 10.57 GB maximum reserved memory usage. Internal testing accuracy was 92.76% and macro F1 93.78%. A separate collection of 176 images from 10 classes had 69.32% accuracy and 71.95% macro F1. External Late Blight recall was 10%, in contrast to 84.62% in internal testing. Due to the unknown benchmark independence and acquisition conditions, this is an indicator of performance difference between collections but not proof of generalization to field conditions.

Keywords: Tomato disease classification · Qwen2.5-VL · QLoRA · Vision-language model · Parameter-efficient fine-tuning

1 Introduction

Image-based plant disease recognition has been studied using CNNs, transfer learning, and broader plant-disease datasets [8, 16, 22, 24]. Tomato-focused work has explored disease and pest recognition, symptom visualization, and

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transformer-based smartphone systems [3, 4, 9], while reviews and datasets highlight the effects of image acquisition and dataset composition on reported performance [2, 5, 12, 18, 20, 21]. Vision Transformers and vision-language models extend image recognition through transformer-based visual representations and language supervision [7,10,15,17,23]. Qwen2.5-VL supports instruction-following multimodal tasks [1], but adapting a multi-billion-parameter model can be computationally demanding. LoRA reduces trainable parameters through low-rank updates, while QLoRA combines low-bit quantization with LoRA to reduce fine-tuning memory requirements [6,11]. Tomato VLM studies and other distribution shift analysis studies indicate the need for assessment of system performance using more than one image set [13, 14, 19].

This study evaluates QLoRA adaptation of Qwen2.5-VL-7B-Instruct for 11 tomato disease and pest classes on one 24-GB GPU. We report results on a frozen 649-image internal test set and a separately stored 176-image TIH-provided evaluation collection. The independence of this benchmark from the private dataset has not been established, and thus it is considered an independent benchmark instead of a proven independent population. The contributions are:

- QLoRA pipeline that uses a frozen vision tower, language-projection adapters, and answer-only supervision.
- Dataset integrity checks that use image validation, SHA-256 exact duplication removal, exclusion of cross-class contradictory duplicates, and stratified splitting.
- Evaluation using internal metrics, benchmark-specific metrics, confusion matrices, accuracy confidence intervals, and analysis of errors of external Late Blight.
- Description of trainable parameters, runtime, GPU memory requirements, and inference latency, along with restrictions on the independence of the benchmark and robustness in the real world.

2 Materials and Methods

2.1 Experimental Overview

Figure 1 provides an overview of data auditing, deterministic splitting, QLoRA modification, validation-based checkpointing choice, and two evaluation phases.

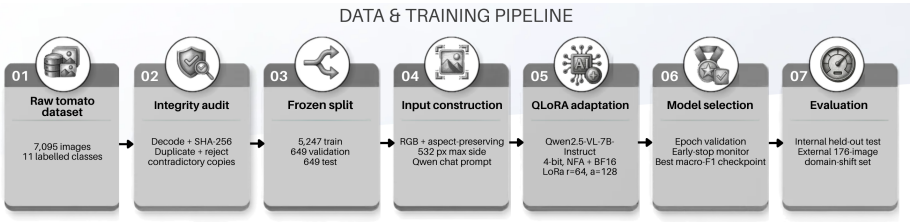


Fig. 1. End-to-end pipeline for fine-tuning and evaluating tomato images.

2.2 Private Tomato Dataset and Class Taxonomy

The original private dataset contained 7,095 labelled images organized into 11 class folders such as Aphid, Early Blight Fruit, Fruit Borer, Healthy Fruit, Healthy Leaf, Late Blight, Leaf Curl, Leaf Miner, Mealy Bugs, Target Leaf Spot, and White Fly. The dataset consists of healthy specimens of tomato plant alongside the classes of diseases and pests. Files were validated against their supported file extensions and were hashed using SHA-256. Byte-identical files within a class were collapsed; hash groups appearing under conflicting labels were excluded. This retained 6,545 images. A deterministic class-stratified 80/10/10 split (seed 20260911) produced 5,247 training, 649 validation, and 649 test images (Table 1). Byte-level deduplication cannot eliminate near-duplicates, and acquisition methods used on the private data set are unknown.

Table 1. Class distribution post-data integrity filters and deterministic splitting.

Class	Train	Validation	Test	Total
Aphid	551	68	68	687
Early Blight Fruit	224	27	27	278
Fruit Borer	346	43	43	432
Healthy Fruit	78	9	9	96
Healthy Leaf	319	39	39	397
Late Blight	1561	195	195	1951
Leaf Curl	424	52	52	528
Leaf Miner	767	95	95	957
Mealy Bugs	320	40	40	400
Target Leaf Spot	264	33	33	330
White Fly	393	48	48	489
Total	5247	649	649	6545

2.3 Image Preprocessing and Prompt Construction

The images have been converted to RGB, resized with maintaining their aspect ratios, with the largest size being 532 pixels, dimension multiples of 28, and bicubic interpolation. The task used constrained image-to-label generation with the instruction: *Classify this tomato plant image into exactly one of these categories: Aphid, Early Blight Fruit, Fruit Borer, Healthy Fruit, Healthy Leaf, Late Blight, Leaf Curl, Leaf Miner, Mealy Bugs, Target Leaf Spot, White Fly. Return only the category name.* The Qwen chat template paired each image and instruction with the exact class label. Prompt tokens were masked (ignore index -100), so loss was computed only on answer tokens; maximum sequence length was 1,024. No synthetic or external training images were added.

2.4 External Benchmark and Evaluation Protocol

The external evaluation dataset was provided by the Technology Innovation Hub for IoT and IoE (TIH), IIT Bombay, and comprised 176 tomato images across 10

classes. Class distribution of the sample set included: Aphid (16), Early Blight Fruit (20), Fruit Borer (20), Healthy (20), Late Blight (20), Leaf Curl (20), Leaf Miner (20), Mealy Bugs (7), Target Leaf Spot (20), and White Fly (13). Because Healthy Fruit was absent, its Healthy folder was mapped to Healthy Leaf. The selected adapter was evaluated without training or checkpoint selection on these images, using the same RGB conversion and resize procedure. The source, acquisition conditions, and independence from the private dataset have not been established; possible exact or near-duplicate overlap therefore cannot be ruled out. The benchmark is not treated as proof of field generalization.

2.5 Qwen2.5-VL QLoRA Configuration

The final training configuration is presented in Table 2. Qwen2.5-VL-7B-Instruct was loaded using 4-bit NF4 quantization with double quantization and BF16 computation. The base model weights remained frozen, while rank-64 LoRA adapters were applied to the language model's `q_proj`, `k_proj`, `v_proj`, and `o_proj` layers. The vision tower was kept frozen and excluded from LoRA adaptation. The adapters used an alpha value of 128, a dropout rate of 0.05, and no bias terms. Overall, 40,370,176 parameters were trainable out of 4,732,627,968 total parameters, accounting for 0.853%. The final checkpoint was selected based on validation macro F1, with validation accuracy and lower validation loss used to break ties. Early stopping used validation loss and had a patience of one evaluation interval. The training process performed 1,908 optimizer steps in 2.909 epochs in 192.7 minutes, with peak reserved GPU memory being 10.57 GB.

2.6 Imbalance-Aware Sampling

Because class frequencies were imbalanced, images were sampled with inverse-square-root weights. For an image in class c with n_c training examples, the weight was $w_c = 1/\sqrt{n_c}$. A weighted sampler drew a training set of the original length with replacement. The square-root correction was selected to reduce majority-class dominance without forcing full inverse-frequency balancing, which would repeatedly expose very small classes such as Healthy Fruit. The sampler was deterministic under the experiment seed and epoch number. Validation and test distributions were not reweighted.

2.7 Optimization and Evaluation

Training used Hugging Face Trainer, paged AdamW 8-bit, learning rate 4×10^{-5}, 3% warmup, cosine decay, zero weight decay, gradient clipping at 1.0, gradient checkpointing, per-device batch size 1, and eight-step gradient accumulation (effective batch size 8). Validation was performed each epoch. Checkpoints were ranked by macro F1, then accuracy and validation loss; early stopping used validation loss with patience one. The environment was Python 3.10, PyTorch 2.6.0+cu124, CUDA 12.4, Transformers 5.16.1, PEFT 0.20.0, and bitsandbytes

Table 2. Final QLoRA training configuration

Component	**Setting**
Base model	Qwen2.5-VL-7B-Instruct
Quantization	4-bit NF4 with double quantization
Compute precision	BF16
Vision tower	Frozen
LoRA target modules	`q_proj`, `k_proj`, `v_proj`, `o_proj`
LoRA rank	64
LoRA alpha	128
LoRA dropout	0.05
LoRA bias	None
Trainable parameters	40,370,176
Total parameters recorded	4,732,627,968
Trainable parameter fraction	Approximately 0.853%
Optimizer	Paged AdamW 8-bit
Learning rate	4×10^{-5}
Weight decay	0
Learning-rate schedule	Cosine decay
Warm-up ratio	3%
Per-device training batch size	1
Gradient accumulation steps	8
Effective batch size	8
Gradient clipping	1.0
Gradient checkpointing	Enabled
Maximum image side	532 pixels, patch-aligned
Maximum sequence length	1,024 tokens
Planned epochs	3
Experiment/split seed	20260911

Note: The reported effective batch size is the configured value for the single-device training run. The recorded run completed 1,908 optimizer steps, corresponding to 2.909 epochs.

0.50.2 on one NVIDIA RTX 4500 Ada GPU (23.99 GB reported memory). The pre-flight checks included software and BF16 compatibility, class mappings, image database, hash splitting, model/loading processors, and evaluation. The three-hour safeguard was enabled, however, the recorded run time was 192.7 minutes. The data that has been obtained does not explain the reason for the exceedance of 180 minutes. One chosen training run was evaluated; hence, run-to-run variance is unknown.

Checkpoint selection was validated, and an evaluation of the 649-image internal test set followed without any further refinement. Metrics included accuracy, macro precision/recall/F1, weighted F1, class-wise metrics, confusion matrices, unparsable outputs, and latency. The separate 176-image collection was evaluated in inference mode without parameter updates. Macro F1 was the primary aggregate metric because of class imbalance; accuracy and weighted F1 are also reported.

3 Results

3.1 Training Efficiency

The run completed 1,908 optimizer updates in 11,562.6 s (192.7 min), with 10.57 GB peak reserved memory and final logged training loss 8.12×10^{-5}. Adapter saving and reloading were verified. Training loss alone is not evidence of generalization; held-out evaluation results follow.

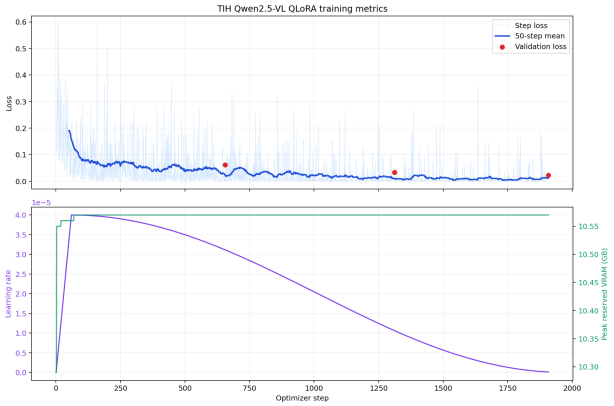


Fig. 2. Training dynamics of the final Qwen2.5-VL QLoRA run, showing recorded step loss, the 50-step moving mean, validation loss, the learning-rate schedule, and peak reserved GPU memory.

3.2 Internal Held-Out Test Performance

On the frozen internal test set, 602/649 images were correct (92.76% accuracy); macro F1 was 93.78%, macro precision 92.71%, macro recall 95.57%, and weighted F1 92.86%. No outputs were unparsable and mean latency was 0.327 s/image. Class-wise F1 ranged from 81.48% (Target Leaf Spot) to 100% (Fruit Borer and Healthy Fruit). Late Blight recall was 84.62% and F1 90.91%; its main confusions were Target Leaf Spot (13), Leaf Curl (8), and Leaf Miner (7) (Fig. 3).

3.3 External Domain-Shift Benchmark

The external collection yielded 122/176 correct (69.32% accuracy), 71.95% macro F1, 76.08% macro precision, 71.84% macro recall, and 69.38% weighted F1. Mean latency was 0.373 s/image, with no unparsable outputs. The 95% Wilson intervals for accuracy were 90.50–94.51% internally and 62.15–75.66% externally. These intervals quantify image-level binomial uncertainty, not training-run variability, dataset-selection bias, or possible overlap. Relative to the internal test, external accuracy was 23.44 percentage points lower and macro F1 was 21.83 points lower. External Late Blight recall was 10% (F1 14.29%); 11 of 20 examples were predicted as Target Leaf Spot. Figure 4 shows the external confusion

matrix, and Fig. 5 and Fig. 6 compare class-wise and aggregate performance. Since acquisition conditions and benchmark independence are unverified, the cause of the difference cannot be determined.

Table 3. Aggregate internal and external evaluation results. Accuracy confidence intervals are 95% Wilson score intervals calculated from the recorded correct and total counts.

Metric	Internal test	External benchmark
Images / classes	649 / 11	176 / 10
Accuracy	92.76% (95% Wilson CI: 90.50–94.51%)	69.32% (95% Wilson CI: 62.15–75.66%)
Macro precision	92.71%	76.08%
Macro recall	95.57%	71.84%
Macro F1	93.78%	71.95%
Weighted F1	92.86%	69.38%
Unparseable predictions	0	0
Mean latency	0.327 s/image	0.373 s/image

3.4 Class-Wise Performance

Table 4 reports internal class-wise precision, recall, and F1 alongside external metrics for the classes present in both collections.

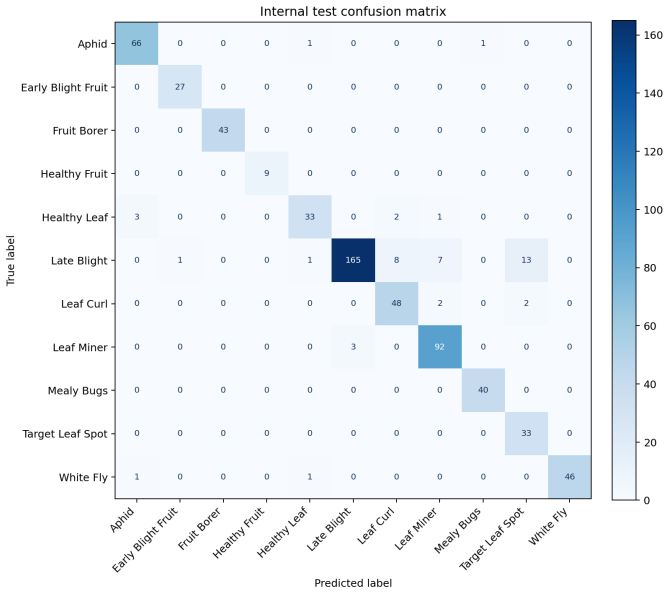


Fig. 3. Confusion matrix on the internal test set.

The class-wise results highlight a substantial performance difference for Late Blight between the internal and external evaluations. On the internal test set,

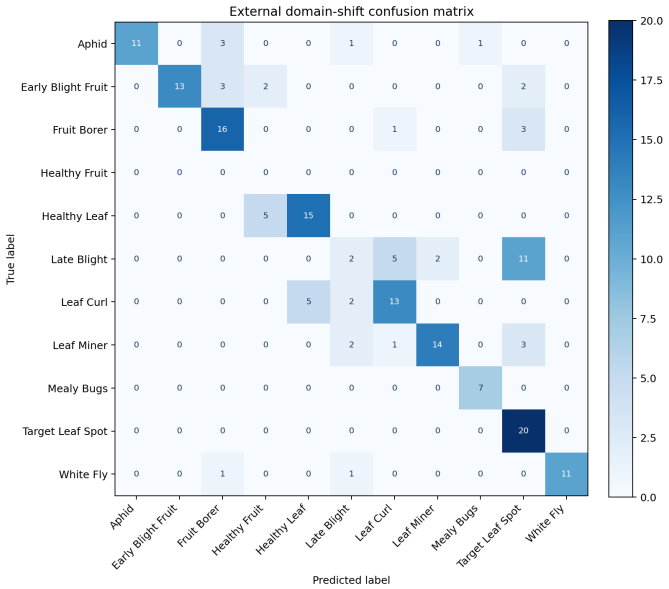


Fig. 4. Confusion matrix on the external benchmark. Healthy Fruit is absent from this collection.

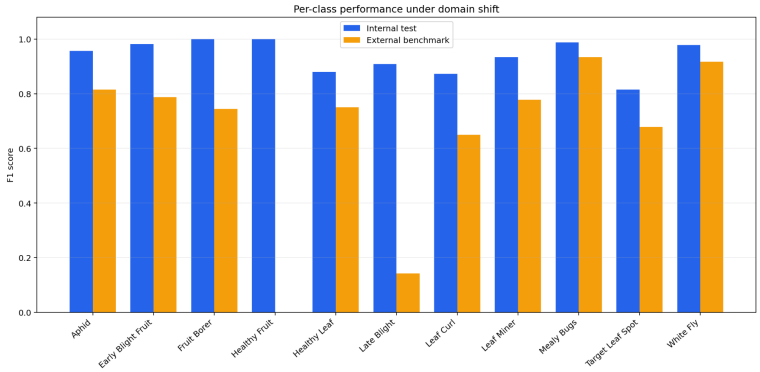


Fig. 5. Per-class F1 comparison between the internal test set and external benchmark.

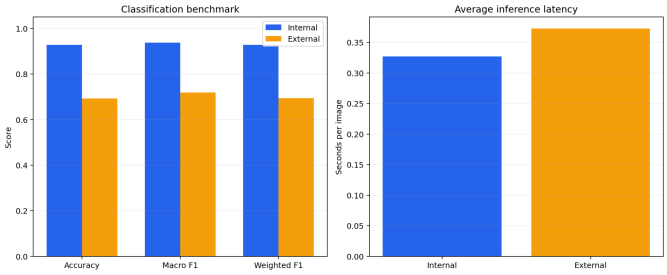


Fig. 6. Benchmark Performance for Internal and External Measures along with Inference Latency.

Table 4. Per-class internal and external evaluation metrics. Values for precision, recall, and F1 are shown in percentages. A dash means the class is missing from the external benchmark set.

Class	Support	Internal Precision (%)	Internal Recall (%)	Internal F1 (%)	External Precision (%)	External Recall (%)	External F1 (%)
Aphid	68	94.29	97.06	95.65	100.00	68.75	81.48
Early Blight Fruit	27	96.43	100.00	98.18	100.00	65.00	78.79
Fruit Borer	43	100.00	100.00	100.00	69.57	80.00	74.42
Healthy Fruit	9	100.00	100.00	100.00	—	—	—
Healthy Leaf	39	91.67	84.62	88.00	75.00	75.00	75.00
Late Blight	195	98.21	84.62	90.91	25.00	10.00	14.29
Leaf Curl	52	82.76	92.31	87.27	65.00	65.00	65.00
Leaf Miner	95	90.20	96.84	93.40	87.50	70.00	77.78
Mealy Bugs	40	97.56	100.00	98.77	87.50	100.00	93.33
Target Leaf Spot	33	68.75	100.00	81.48	51.28	100.00	67.80
White Fly	48	100.00	95.83	97.87	100.00	84.62	91.67

Late Blight achieved 84.62% recall and an F1-score of 90.91%, whereas the external evaluation yielded 10.00% recall and an F1-score of 14.29%. Of the 20 external Late Blight images, 11 were classified as Target Leaf Spot. This confusion identifies Late Blight as the principal class-level error pattern in the external evaluation and indicates that the overall external metrics alone do not fully describe the model's performance across disease categories.

4 Discussion

From the results, QLoRA trained Qwen2.5-VL-7B-Instruct on the 11-class problem using 0.853% of the model's parameters. On the training set, Qwen2.5-VL-7B-Instruct demonstrated an internal macro F1 of 93.78%, while the externally stored evaluation collection showed a macro F1 of 71.95%. It is evident that there is a discrepancy between the performance of this model in both cases, especially the Late Blight F1 score, which fell from 90.91% to 14.29%. Hence, an internally consistent performance metric is not sufficient to characterize the performance on other collections. There are 20 Late Blight images outside the collection, of which 11 are classified as Target Leaf Spot. That shows an example of the observed confusion pattern, but there are no grounds for stating from the available data whether that is due to differences in background, camera, lighting, symptom severity, or annotation. There is an imbalance in the training distribution, as Late Blight is the biggest class while Healthy Fruit is the smallest class. Inverse-square-root sampling mitigates class dominance but does not resolve the visual similarity.

4.1 Limitations

The private dataset's acquisition protocol is unavailable, exact-byte deduplication does not detect all near-duplicates, and the external collection's provenance, acquisition conditions, and independence from training data are not established. It contains 176 images across 10 classes; Healthy Fruit is absent and Healthy was mapped to Healthy Leaf. Only one training run was evaluated, so run-to-run variability is unknown. Wilson intervals quantify image-level accuracy uncertainty only. Hyperparameters were not assessed through an extensive search. The model performs image-level classification only: it does not localize lesions, estimate severity, track progression, or recommend treatment.

5 Practical and Societal Implications

The pipeline demonstrates QLoRA adaptation on one 24-GB GPU, but practical agricultural use remains prospective. Field trials, farmer evaluation, crop-yield measurements, and pesticide or economic impact analyses were not conducted; therefore, no direct societal or economic benefit is established. The model may be investigated as a decision-support component, but should not substitute for expert diagnosis. Future evaluation should use documented, independently acquired collections with broader acquisition conditions and repeated training runs, without using external test images for training or checkpoint selection.

6 Conclusion

QLoRA fine-tuning adapted Qwen2.5-VL-7B-Instruct to 11 tomato disease and pest classes using 40.37 million trainable parameters and 10.57 GB peak reserved GPU memory. The selected adapter achieved 92.76% internal accuracy and 93.78% macro F1, compared with 69.32% accuracy and 71.95% macro F1 on a separately stored 176-image evaluation collection. External Late Blight recall was only 10%, with 11 of 20 images confused with Target Leaf Spot. Because dataset independence and acquisition conditions remain unverified and repeated runs were not conducted, these findings demonstrate a performance gap between the evaluated collections rather than established field generalization. Broader independently documented evaluation is required before deployment claims can be made.

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